

An example of solving with radicals.

1. Find all real solutions of $\sqrt{x+10} + \sqrt{x-9} = 19$.

This problem is probably easiest to tackle by isolating one of the radicals, squaring and simplifying; and then isolating the other radical.

$$\begin{aligned}
 \sqrt{x+10} + \sqrt{x-9} &= 19 \\
 \Rightarrow \sqrt{x+10} &= 19 - \sqrt{x-9} \\
 \Rightarrow (\sqrt{x+10})^2 &= (19 - \sqrt{x-9})^2 \\
 \Rightarrow x+10 &= 361 - 38\sqrt{x-9} + x-9 \\
 \Rightarrow -342 &= -38\sqrt{x-9} \\
 (-342)^2 &= (-38\sqrt{x-9})^2 \\
 116964 &= 1444(x-9) \\
 81 &= x-9 \\
 x &= 90
 \end{aligned}$$

While a little too complicated for an exam, this problem is good algebra practice. The concept of isolating the radical so it can be squared is key.

Problem 12 from section 3.4: The William, Nancy, and Edgar distance problem.

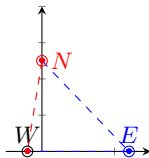
If they all start at the same time:

2. William, Nancy, and Edgar met at the lodge. At 8 AM William began hiking at 2 mph, Nancy began hiking north at 5 mph, and Edgar went East on a three-wheeler at 12 mph. At what time was the distance between Nancy and Edgar 20 miles greater than the distance between Nancy and William?

We need to solve the inequality

$$\text{dist}(N, E) > \text{dist}(N, W) + 20 \quad (0.1)$$

Where N = Nancy's position, E = Edgar's position, and W = William's position as (x, y) coordinates with the cabin at the origin (point $(0, 0)$). See the figure:



$N = (0, 5t)$, $E = (12t, 0)$, and $W = (-2t, 0)$. The "—" for William nearly denotes his direction (it is not really necessary). Each point is given with respect to time after 8 AM. At $t = 0$, they are all at the cabin; $(0, 0)$. They all move away from the cabin according to their speed and direction. The picture shows $t = 1$.

$\text{dist}((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$, so:

- $\text{dist}(N, W) = \sqrt{(0 - (-2t))^2 + (5t - 0)^2} = \sqrt{4t^2 + 25t^2} = \sqrt{29t^2} = \sqrt{29}t$

- $\text{dist}(N, E) = \sqrt{(0 - 12t)^2 + (5t - 0)^2} = \sqrt{144t^2 + 25t^2} = \sqrt{169t^2} = 13t$

Plugging this into equation 0.1, we have:

$$\begin{aligned}
 13t &> \sqrt{29}t + 20 \\
 13t - \sqrt{29}t &> 20 \\
 (13 - \sqrt{29})t &> 20 \\
 t &> \frac{20}{13 - \sqrt{29}} \approx 2.63 \text{ hrs after 8AM}
 \end{aligned}$$

So about $8 + 2.63 = 10.63$ AM or $10 : (.63) \cdot 60 = 10 : 38$ AM

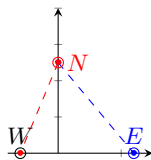
If William gets a head start:

3. William, Nancy, and Edgar met at the lodge at 8 AM, and William began hiking at 2 mph. At 10 AM, Nancy began hiking north at 5 mph and Edgar went East on a three-wheeler at 12 mph. At what time was the distance between Nancy and Edgar 20 miles greater than the distance between Nancy and William?

We need to solve the inequality

$$\text{dist}(N, E) > \text{dist}(N, W) + 20 \quad (0.2)$$

Where N = Nancy's position, E = Edgar's position, and W = William's position as (x, y) coordinates with the cabin at the origin (point $(0, 0)$). See the figure:



$N = (0, 5t)$, $E = (12t, 0)$, and $W = (-2t - 4, 0)$. William starts 2 hours earlier hence the $-2 \cdot 2 = -4$ (the negative denotes direction; it turns out to be unnecessary in this problem). Each point is given with respect to time after 10 AM. At $t = 0$, N and E they are at the cabin, $(0, 0)$; W is 4 miles west of the cabin, $(-4, 0)$. They all move away from the cabin according to their speed and direction. The picture shows $t = 1$.

$\text{dist}((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$, so:

- $\text{dist}(N, W) = \sqrt{(0 - (-2t - 4))^2 + (5t - 0)^2} = \sqrt{4t^2 + 16t + 16 + 25t^2} = \sqrt{29t^2 + 16t + 4}$
- $\text{dist}(N, E) = \sqrt{(0 - 12t)^2 + (5t - 0)^2} = \sqrt{144t^2 + 25t^2} = \sqrt{169t^2} = 13t$

Plugging this into equation 0.2, we have:

$$\begin{aligned} 13t &> \sqrt{29t^2 + 16t + 4} + 20 \\ 13t - 20 &> \sqrt{29t^2 + 16t + 4} \\ (13t - 20)^2 &> \left(\sqrt{29t^2 + 16t + 4}\right)^2 \\ 169t^2 - 520t + 400 &> 29t^2 + 16t + 4 \\ 140t^2 - 536t + 396 &> 0 \\ 4(t - 1)(35t - 99) &> 0 \end{aligned} \quad (0.3)$$

So we need to solve the quadratic inequality, $4(t - 1)(35t - 99) = 0$. So $t = 1$ or $t = \frac{99}{35} \approx 2.829$. But $t = 1 \Rightarrow 13(1) - 20 = -7$ which is not a solution for the inequality 0.3 above because the radical cannot be less than 0. So then $t \approx 2.829$ hrs.

So the solution is approximately $10 + 2.829 = 12.83$ PM or 12 : 50 PM.

While a good problem, it is difficult because there are so many little details to mess up!

Note that the negative used for W 's position was unnecessary because it got squared. However if you wanted to find the distance between W and E it would be needed. the distance between points could have been found using Pythagoras's Theorem, but this theorem is really just a distance formula.